

Algorithms, AI & Data Science II

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Notes:

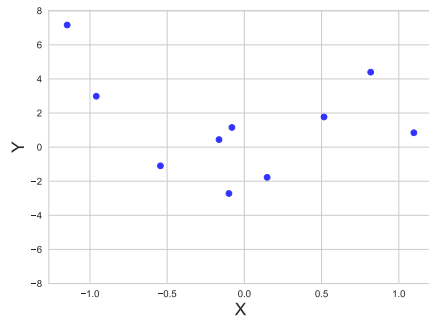
- **Lecture L12: Testing Hypotheses**

18.06.2025

- **Educational objective:** We explain how we can use data to test statistical propositions. We apply this approach to test for the presence of linear relationships in data and discuss basic regression diagnostic techniques.
 - Statistical Hypothesis Testing
 - Testing Linear Relationships
 - Regression Diagnostics
- **Handout version:** 2025/06/03 14:59:00

Motivation

- ▶ we can use frequentist or Bayesian methods to infer **plausible parameters of statistical models** given data
- ▶ example: slope/intercept of **linear model** that explains dependent variable Y based on independent variable X
- ▶ how do we tell **whether a linear relationship exists** in the data?
- ▶ how can we use data to give yes/no answers to **statistical propositions**?
- ▶ important example of **inductive statistical reasoning** and key challenge in **statistical inference**
- ▶ basis for **empirical sciences** and one of the foundational methods in **data science**



samples from bivariate random variable (X, Y)

example proposition: there is a linear relationship between X and Y

Notes:

Philosophical foundations of science

► **epistemology**: how can we extract knowledge from observations?

classical rationalism



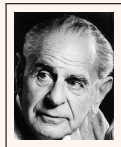
Gottfried Wilhelm Leibniz

1646 – 1716

science as objective search for **infallible truths**
based on deductive reasoning

The senses, although [...] necessary for all our actual knowledge, are not sufficient to give us the whole of it, since the senses never give anything but instances, that is to say particular or individual truths. Now all the instances which confirm a general truth, however numerous they may be, are not sufficient to establish the universal necessity of this same truth [...] From which it appears that necessary truths, such as we find in pure mathematics [...] must have **principles whose proof does not depend on instances**, nor consequently on the testimony of the senses [...] → Gottfried Wilhelm Leibniz: *New Essays on Human Understanding*, 1704

critical rationalism



Sir Karl Popper

1902 – 1994

scientific theories must make **predictions that can be falsified**
based on observations and experiments

“[Eine Einstellung] die zugibt, dass ich mich irren kann, dass du recht haben kannst und dass wir zusammen vielleicht der Wahrheit auf die Spur kommen werden.”

→ Karl Popper: *Die offene Gesellschaft und Ihre Feinde I*, 1944

Notes:

- Within philosophy, the core field of epistemology (a combination of the Greek words episteme (knowledge) and logos (science)) addresses the question how we can extract knowledge from observations. In the introductory slides, I gave a quick overview of the scientific method, which uses models or theories that need to make predictions, which can then be invalidated in experiments or further observations. This approach to science is called critical rationalism.
- Notably, this is not the only approach that has been considered to generate knowledge. In fact, the idea that we want to invalidate theories based on observations or data is a fairly recent perspective in the long history of science. In previous centuries, the dominant approach was based on a rationalist view, which sees science as objective (and deductive) search for infallible truths that should ideally be independent of subjective experiences (and thus observations).
- A major contribution of critical rationalism is that it explicitly calls for **theories that can be proven wrong by counterexamples**. This addresses the problem, that —apart from mathematical theorems— we can typically not prove that a theory is correct. In the critical rationalist approach, a theory becomes more and more plausible the more attempts to falsify it have failed.

Science and falsifiability



Richard Feynman

1918 – 1988

“We never are right, we can only be sure we are wrong.”

→ Richard Feynman

image credit: Tamiko Thiel 1984 - OTRS communication from photographer, CC BY-SA 3.0



Wolfgang Pauli

1900 – 1958

“Das ist nicht nur nicht richtig; es ist nicht einmal falsch!”

→ Wolfgang Pauli

image credit: Nobel foundation, public domain

Notes:

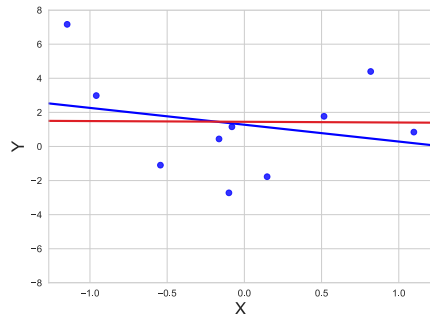
- This introduces an important requirement for a theory or model to be considered a scientific theory: it must make predictions that we can possibly falsify in an experiment. Theories that, even though they may explain a phenomenon, do not make such predictions, are not considered scientific. In fact, the fact that a scientific theory must allow it to be falsified is what separates scientific theories from many so-called “conspiracy theories”. A feature of the latter is that we can never find any evidence that contradicts the theory.

Testing hypotheses

- ▶ presence of relationship is a **hypothesis**, i.e. a theory about the data that we try to **falsify**
- ▶ we consider **null hypothesis** (H_0) as well as **alternative hypothesis** (H_1)

example

- ▶ H_0 : there is no linear relationship in our data ($\beta_1 = 0$)
 - ▶ H_1 : there is a linear relationship in our data ($\beta_1 \neq 0$)
-
- ▶ we test whether we can **reject null hypothesis against alternative**, i.e. whether our confidence in alternative is larger than in null hypothesis
 - ▶ deductive vs. inductive reasoning: we never prove a hypothesis (or theory), but may **fail to falsify** it



“absence of evidence is not evidence of absence”

Notes:

Statistical hypothesis testing

hypothesis testing procedure

1. formulate **null hypothesis** H_0 and **alternative hypothesis** H_1
2. collect data
3. compute **test statistic** t that summarizes data such that H_0 and H_1 are distinguishable
4. calculate $p := P(T \geq t)$, where T is a **random variable assuming values of t under H_0**
5. choose **significance threshold** α

► $p \geq \alpha$

- “Our observation is consistent with H_0 .”
- “We have no significant evidence against H_0 .”
- “We cannot reject H_0 in favor of H_1 .”

► $p < \alpha$

- “We reject H_0 in favor of H_1 .”

p -value quantifies our belief in null hypothesis H_0 compared to alternative H_1 .
 p -value is **NOT** the probability that the null hypothesis is true!



Ronald Fisher

1890 – 1962

image credit: Wikipedia, CC-BY-SA

Notes:

- We introduce techniques that help us to formally test hypotheses using statistical methods. We start our discussion with a rather uncritical view of statistical hypothesis testing based on p -values. In the next lecture we will add a discussion of the “dos and don’ts” of statistical hypothesis testing in high-dimensional data.
- The basic idea is to compute a test statistic t and then compute a probability p to observe a test statistic that is as least as “extreme” as t , if the null hypothesis was true.
- If this probability is smaller than a chosen significance threshold α , we reject H_0 otherwise we fail to reject it. We need to be careful with our wording: **we never accept the null hypothesis**. We simply fail to reject it compared to a specific alternative hypothesis - and we may actually succeed rejecting it if we had compared it against another alternative. **We also never prove or accept the alternative hypothesis**. We may reject the null hypothesis in favour of an alternative hypothesis, which does not rule out that another alternative hypothesis might give an even better explanation.
- The p -value should be interpreted as the level of confidence regarding two competing hypotheses. Any dichotomization (i.e. reject/accept, true/false) must be done in a very careful way. We will see in a moment how we can choose the threshold α .
- It is important to highlight that the **p -value is NOT the probability that the null hypothesis is true**, which is a common misunderstanding. The p -value is rather the probability to observe data where the chosen test statistic t is at least as extreme as what we observed, if the null hypothesis was true.

Example: Is the game fair?

coin tossing game

you are offered a bet on a coin toss with the following conditions:

- ▶ the stake is EUR 100
- ▶ the other party wins when the coin shows tails
- ▶ you win EUR 500 when the coin shows heads
- ▶ the other party claims that the coin is fair, i.e. $q := P(\text{heads}) = \frac{1}{2}$

You lose! Can you conclusively argue that the coin was manipulated?

scientific approach

1. we **formulate null and alternative hypothesis**
 - ▶ H_0 : coin is fair ($q = \frac{1}{2}$)
 - ▶ H_1 : coin is biased against you ($q < \frac{1}{2}$)
2. we **collect data** on $n = 20$ coin tosses
3. we **compute test statistic** $t = 18 \times \text{tails}$
4. $T \geq 18$ is more “extreme” outcome under H_0 compared to H_1



image credit: Ivan Radic, Flickr, CC-BY 2.0

Notes:

- To better understand the advantages and limitations of statistical hypothesis testing, let us consider a concrete example. We assume that you pass by a dark street corner in Wuppertal and you are approached by two persons. They offer you a game of a coin toss, where – obviously – they provide the coin. They claim that the coin is fair and offer betting odds of 5/1 for heads. In other words, you bet EUR 100 on heads and you get EUR 500 if the coin shows heads. If the coin shows tails, you lose your money. Since the coin is fair you obviously accept this very favourable bet. Your “business partners” throw the coin, it shows tails and you lose your money. After rethinking the whole situation you become a bit suspicious that the coin was manipulated. How can you show this based on statistical methods?
- Taking a scientific approach in our investigation, we first formulate a hypothesis. Following the conservative principle “in dubio pro reo” (in doubt, for the defendant, cf. critical rationalism!), our null hypothesis is that the game was actually fair, i.e. the probability q of heads is $\frac{1}{2}$. Our alternative hypothesis is that the coin is biased against us, i.e. $q < \frac{1}{2}$.
- We now try to falsify our theory that the bet was actually fair. For this, we need data. We are lucky, and peeking around the corner we are able to observe the outcomes of 20 more bets offered to other unsuspecting people. We observe that the coin shows tails 18 times and two times heads. We define our test statistic to be the number of tails, i.e. $t = 18$.

Group Exercise 12-01

- Assume an observation of $18 \times$ tails in $n = 20$ tosses of a coin with unknown q . Calculate the p -value of the null hypothesis that the coin is fair ($q = \frac{1}{2}$) against the **alternative hypothesis that the coin is biased against you** ($q < \frac{1}{2}$).
- Give a probabilistic interpretation of the resulting p -value.

for alternative hypothesis $H_1 (q < \frac{1}{2})$ the test statistic is $t = 18 \times$ tails. Considering the random variable T assuming the number of tails under the null hypothesis that the coin is fair, we have: p -value of **null hypothesis** $H_0 (q = \frac{1}{2})$ against **alternative** $H_1 (q < \frac{1}{2})$

$$p := P_{H_0}(T \geq 18) = \sum_{k=18}^{20} \binom{20}{k} \frac{1}{2^k} \frac{1}{2^{20-k}} = 2^{-20} \cdot \sum_{k=18}^{20} \binom{20}{k} = 2^{-20} (20 + 1 + 190) \approx 0.0002$$

interpretation

If the coin was fair (i.e. under H_0), the probability to observe at random a test statistic that is **at least as extreme as** 18 tails is 0.02 %. We reject H_0 in favor of H_1 (e.g. at significance threshold $\alpha = 0.05$).

Notes:

- We can calculate the p -value of our null hypothesis against the alternative hypothesis H_1 based on the binomial distribution. Under H_0 the test statistic t follows a binomial distribution with probability 0.5. We use the probability $P(k)$ to observe k times heads under the null hypothesis $q = \frac{1}{2}$, sum it for $k \in [18, 19, 20]$ and obtain $P(T \geq 18)$.
- Not surprisingly, we find that the probability to observe 18 or more times heads under the null hypothesis is extremely small. Hence, the p -value of H_0 against the alternative H_1 is very small.
- It may seem that this calculation is independent of the alternative hypothesis, but this is actually not true. The choice of the alternative hypothesis influences our test statistic t and thus the p -value. To better understand this, consider the case that you test H_0 against the alternative hypothesis $q > \frac{1}{2}$, i.e. our alternative would now be that the game is manipulated in our favour! In this case, the direction of what is considered an “extreme” outcome of the test statistic changes.
- We now use the number of heads as the test statistic t and calculate the p -value in the same way as above. We obtain a large p -value close to 1 (which is just the inverse of the p -value of the opposite alternative hypothesis).
- This simple example shows that (i) it is crucial to choose the “correct” alternative hypothesis considering what you want to show, and (ii) that you have to choose a meaningful test statistic given your choice of the alternative hypothesis.

Choosing a significance threshold

- choice of **significance threshold** α determines whether we **reject null hypothesis** H_0 in favour of alternative H_1

we reject H_0 in favour of H_1 iff $p < \alpha$

- we can make two different errors

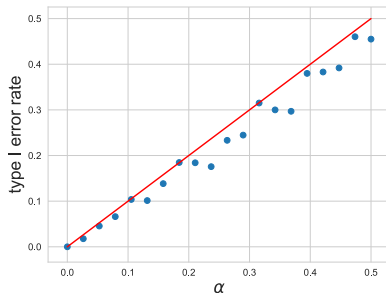
	reject H_0	do not reject H_0
H_0 is true	type I error	OK
H_0 is false	OK	type II error

choice of α ensures that **type I error rate is** α , i.e. we wrongly reject H_0 at rate α

example

type I error rate for 1000 tests of null hypothesis H_0 ($q = \frac{1}{2}$) against alternative H_1 ($q < \frac{1}{2}$) for significance thresholds α

data: 100 coin tosses with $q = \frac{1}{2}$



blue dots: empirical type I error rate
red line: type I error rate α

Notes:

- We have found a p -value of 0.0002. We now want to take a decision whether we want to call the police or not. How strong is our evidence? This raises the crucial question about (i) the interpretation of the p -value and (ii) how we choose the significance threshold α below which we reject the null hypothesis in favour of the alternative hypothesis.
- We repeat that the p -value is NOT the probability that the null hypothesis is true, i.e. it is **not** the probability that the coin is fair! But there nevertheless is a natural interpretation in terms of the error that we may make if we take a decision based on p . What errors could we possibly make? The null hypothesis can either be true or not, which means that we either (i) wrongly reject it if it is actually true, or (ii) wrongly not reject it if it is actually wrong. In the other two cases of the four possible outcomes, we do not make a mistake. We refer to the first error as “type I error” and we refer to the second error as “type II error”.
- The p -value is defined as the probability to see a test statistic as extreme as in our data at random, if the null hypothesis were true. This implies that, if we reject the null hypothesis because the value is below α , the probability to make a type I error in n tests is α . In our example, this means that if we observe 20 coin tosses 10000 times, we expect to erroneously call the police two times.
- The plot above shows that this holds. Here we have repeated our coin toss experiment 10000 times, calculating the type I error rate (i.e. the fraction of wrongly rejected null hypotheses in 10000 experiments) for different values of the significance threshold α used in the hypothesis test.

Group Exercise 12-02

- ▶ Assume an observation of $18 \times$ tails in $n = 20$ tosses of a coin with unknown q . Calculate the p -value of the null hypothesis that the coin is fair ($q = \frac{1}{2}$) against the **alternative hypothesis that the coin is biased in your favour** ($q > \frac{1}{2}$).
- ▶ Give a probabilistic interpretation of the resulting p -value.

for **alternative hypothesis** H_2 ($q > \frac{1}{2}$) the test statistic is $t := 2 \times \text{heads}$ ($T \geq 2$ is more “extreme” under H_0 compared to H_2). Considering the random variable T assuming the number of heads under the null hypothesis that the coin is fair, we have:

$$p := P_{H_0}(T \geq 2) = 1 - P(T < 2) = 1 - \sum_{k=0}^1 \binom{20}{k} \frac{1}{2^k} \frac{1}{2^{20-k}} = 1 - 2^{-20} \cdot 20 \approx 0.99998$$

interpretation

If the coin was fair, the probability to observe at random a test statistic that is **at least as extreme as** two heads is 99.998 %. We cannot reject H_0 in favor of H_2 .

Notes:

- We can now calculate the p -value of our null hypothesis against the alternative hypothesis H_1 based on the binomial distribution. Under H_0 the test statistic t follows a binomial distribution with probability 0.5. We use the probability $P(k)$ to observe k times heads under the null hypothesis $q = \frac{1}{2}$, sum it for $k \in [18, 19, 20]$ and obtain $P(T \geq 18)$.
- Not surprisingly, we find that the probability to observe 18 or more times heads under the null hypothesis is extremely small. Hence, the p -value of H_0 against the alternative H_1 is very small.
- It may seem that this calculation is independent of the alternative hypothesis, but this is actually not true. The choice of the alternative hypothesis influences our test statistic t and thus the p -value. To better understand this, consider the case that you test H_0 against the alternative hypothesis $q > \frac{1}{2}$, i.e. our alternative would now be that the game is manipulated in our favour! In this case, the direction of what is considered an “extreme” outcome of the test statistic changes.
- We now use the number of heads as the test statistic t and calculate the p -value in the same way as above. We obtain a large p -value close to 1 (which is just the inverse of the p -value of the opposite alternative hypothesis).
- This simple example shows that (i) it is crucial to choose the “correct” alternative hypothesis considering what you want to show, and (ii) that you have to choose a meaningful test statistic given your choice of the alternative hypothesis.

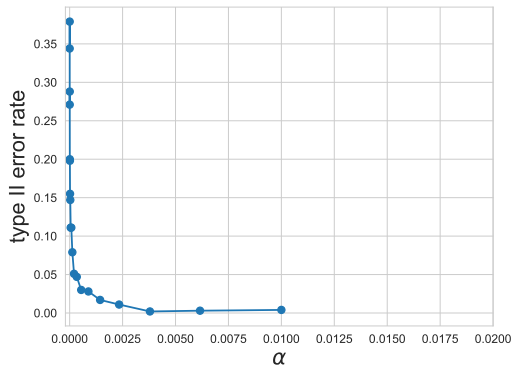
Type II error rate

1/2

type II error rate for H_0 against H_1

type II error rate for 1000 tests of null hypothesis H_0 ($q = \frac{1}{2}$) against alternative hypothesis H_1 ($q > \frac{1}{2}$) for different significance thresholds α

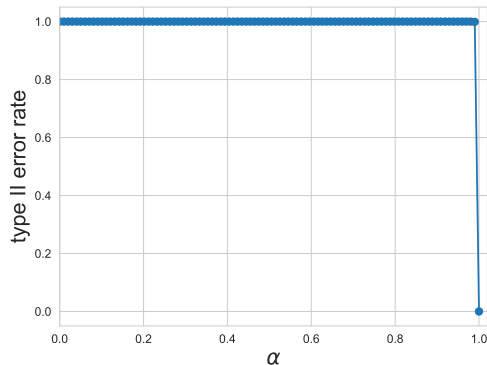
data: 100 coin tosses with $q = \frac{3}{4}$.



type II error rate for H_0 against H_2

type II error rate for 1000 tests of null hypothesis H_0 ($q = \frac{1}{2}$) against alternative hypothesis H_2 ($q < \frac{1}{2}$) for different significance thresholds α

data: 100 coin tosses with $q = \frac{3}{4}$



Notes:

- In our specific example, controlling the type I error rate limits the probability to wrongly accuse our “business partners”. In scientific terms, the significance threshold limits the probability to erroneously falsify a theory. But what we can say about the type II error rate? We can always get the type I error to zero simply by never rejecting the null hypothesis. In our example: If we never go to the police we can never make a wrong accusation. Obviously this does not solve our problem.
- We should rather minimise the type II error rate while controlling for the type I error rate (which is determined by the significance threshold α). Unfortunately, we cannot make general statements as the type II error rate depends on:
 1. significance threshold α : if we were to use $\alpha = 1$, we never do not reject the null hypothesis wrongly, i.e. we have zero type II errors but the type I error is one, i.e. we always wrongly reject H_0 .
 2. choice of alternative hypothesis and associated test statistic: if we choose a non-sensical alternative hypothesis, the p -value still controls the type I error rate, but our test loses any discriminative power, i.e. the type II error rate is close to one for any threshold $\alpha < 1$ (see example on the right). We can see this in the example above: Using the “correct” alternative hypothesis for our scenario (left), the type II error rate quickly drops to values close to zero as we increase α (and thus allow for a larger type I error).
 3. variance of data and sample size: For small data sets or data with high variance we cannot reject the null hypothesis at small thresholds, i.e. our type II error rate tends to be high for small α .

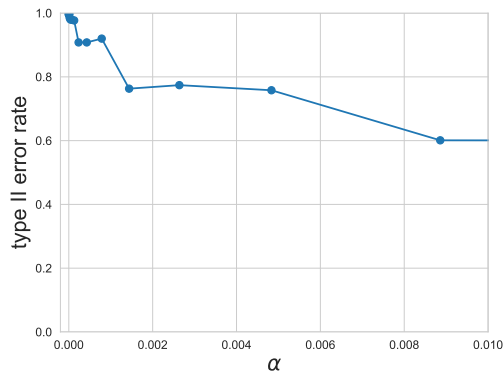
Type II error rate

2/2

type II error rate for H_0 against H_1

type II error rate for 1000 tests of null hypothesis H_0 ($q = \frac{1}{2}$) against alternative hypothesis H_1 ($q > \frac{1}{2}$) for different significance thresholds α

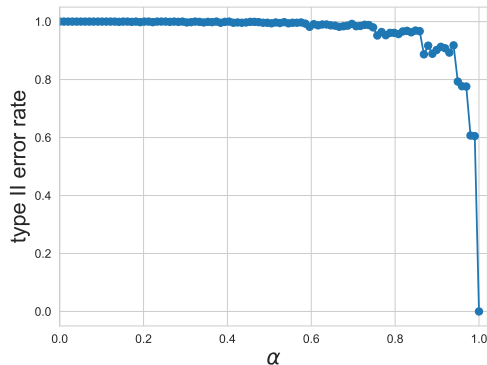
data: 20 coin tosses with $q = \frac{3}{4}$.



type II error rate for H_0 against H_2

type II error rate for 1000 tests of null hypothesis H_0 ($q = \frac{1}{2}$) against alternative hypothesis H_2 ($q < \frac{1}{2}$) for different significance thresholds α

data: 20 coin tosses with $q = \frac{3}{4}$

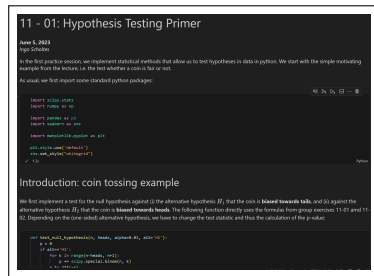


Notes:

- In the examples above, we have changed the number of observed coin tosses used in our hypothesis test from 100 to 20. With less data, it is more difficult to correctly reject the null hypothesis against the alternative. Again, we find that the type II error rate additionally depends on the choice of the alternative hypothesis.

Practice Session

- ▶ we implement **statistical hypothesis testing** in python
- ▶ we calculate the **p -value of a null hypothesis** for different alternative hypotheses
- ▶ we use simulations to explore the **type-I and type-II error rate** for different sample sizes and alternative hypotheses



practice session

see notebook 12-01 in gitlab repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

Notes:

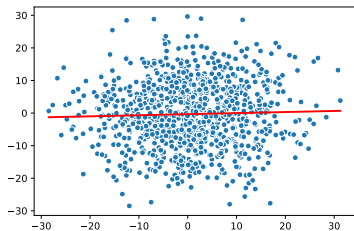
Testing linear relationships

given samples from bivariate random variable (X, Y) , how can we test whether a linear relationship exists between X and Y ?

example 1

1000 realizations of (X, Y) with $X \sim \text{Norm}(0, 10)$ and

► $Y \sim 0.05 \cdot X + \text{Norm}(0, 10)$



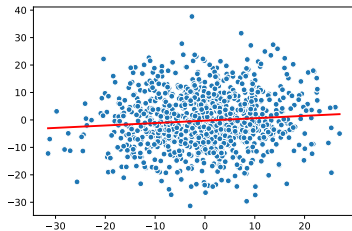
$$\hat{\beta}_1 = 0.088$$

assume $\beta_1 = 0 \Rightarrow$ **type II error**

example 2

1000 realizations of (X, Y) with $X \sim \text{Norm}(0, 10)$ and

► $Y \sim \text{Norm}(0, 10)$, i.e. X and Y are **independent**



$$\hat{\beta}_1 = 0.033$$

assume $\beta_1 > 0 \Rightarrow$ **type I error**

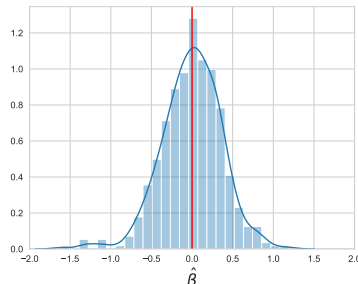
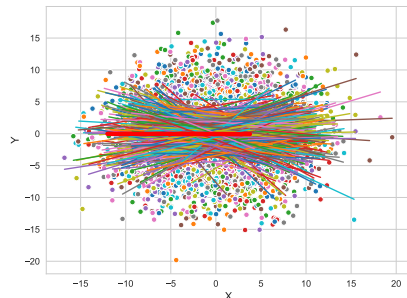
Notes:

- Now that we have covered basics of statistical hypothesis testing, we can come back to our motivating example of linear regression. Let us consider the two data sets above, along with the two estimates of the underlying linear relationship.
- In the first example (shown on the left), there is a linear relationship in the data (see data generation process), but it is weak and the variance in the data is relatively large. We find an estimate for the slope that is slight larger than zero. If we were to assume that there is no relation between X and Y (i.e. we say that $\beta_1 = 0$) we would not reject the null hypothesis despite the fact that it is not true (the true slope is larger than zero), i.e. we would make a type II error.
- In the second example (shown on the right), we again obtain a slope slightly larger than zero, despite there being no relationship whatsoever in the data (see the data generation procedure). If we assume that there is a linear relationship between X and Y (i.e. we say that $\beta_1 > 0$) we would wrongly reject the null hypothesis despite it being true (the true slope is zero), i.e. we would make a type I error.
- In the following, we want to extend statistical hypothesis testing to linear regression models. We make use of the standard error, i.e. the estimation of the standard deviation for the residuals that we have introduced in the previous lecture.

Distribution of estimated slopes

- ▶ consider 100 random samples of $n = 10$ realizations of bivariate RV (X, Y) where X and Y are independent, i.e. there is **no linear relationship** ($\beta_1 = 0$)
- ▶ for each sample of $n = 10$ realizations infer $\hat{\beta}_1$ of linear model, i.e. we obtain **distribution of estimates**
- ▶ due to CLT, estimates are **normally distributed around true value of $\beta_1 = 0$** , i.e. $\hat{\beta}_1 \sim \text{Norm}(0, \sigma)$ → L09
- ▶ in data of size n with normally distributed residuals ϵ_i we can use **standard error** to estimate standard deviation of fitted parameter $\hat{\beta}_1$ from true mean β_1 → exercise sheet

$$\text{SE} = \sqrt{\frac{\frac{\text{Var}(\epsilon_i)}{\text{Var}(X)}}{n - 2}}$$



Notes:

- To test the hypothesis whether a relationship exists or not, we first consider a simple example. The figure above shows the estimates (coloured fit lines) in random samples of a data set that does not contain a linear relationship. Under the assumptions of least squares linear regression (i.e. normally distributed, independent, and homoskedastic residuals) the mean of those estimates approaches the true slope (zero) as we sample more data points.
- Moreover, we can use the variance of the residuals in our data to calculate a standard error of our estimate, which we can also use to calculate a confidence interval of our estimate, i.e. we know that in approx. 95 out of 100 times our estimated parameters will be within two standard errors around the true relationship in the data.
- But how can we dichotomize our answer to the question whether a linear relationship exists or not?
- Intuitively, we should be able to compute a p -value for the hypothesis that the true slope is different from zero based on (i) the distance of the estimated slope $\hat{\beta}_1$ from zero, (ii) the standard error of the estimate, and (iii) the number of data points and the variance in our data. After all, due to the normal distribution of the parameter estimates, if there is no relationship and the standard error of our estimate is small it is unlikely that the estimated slope is far from zero.

Student's t-distribution

Student's t-distribution

- ▶ continuous distribution on $\mathbb{R}$

$$\text{Stud}(t; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

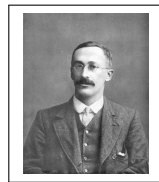
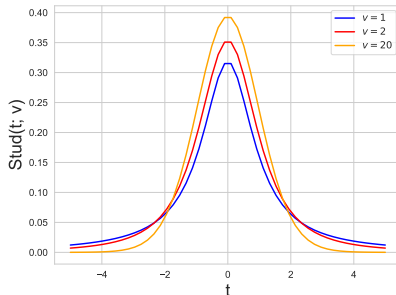
where $\nu := n - 1$ are degrees of freedom and Γ is the gamma function.

- ▶ mean: 0 for $\nu > 1$
- ▶ variance: $\frac{\nu}{\nu-2}$ for $\nu > 2$
- ▶ for $\text{Stud}(t; \nu) \rightarrow \text{Norm}(0, 1)$ for $\nu \rightarrow \infty$

example

For n values x_i drawn from $\text{Norm}(\mu, \sigma)$ with $\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i$ and SE standard error of estimate $\bar{x}$ of true mean μ we have

$$t := \frac{\bar{x} - \mu}{SE} \sim \text{Stud}(t; n - 1)$$



William Sealy Gosset (aka “Student”)

1876 – 1937

Notes:

- We use the fact that the estimates of a linear model are normally distributed around the true value, with a standard deviation that we can estimate based on the residuals. We can now use a fairly general statistical hypothesis testing procedure for that we introduce in the following:
- Let us first define Student's t -distribution, a continuous distribution on $\mathbb{R}$ that is the foundation for the so-called t -test. The parameter ν is also called normality parameter, because for larger ν the t -distribution approaches the standard normal distribution $\text{Norm}(0, 1)$.
- The t -distribution has an intriguing property: Consider a sample of n values drawn from a normal distribution with unknown mean and variance. We can compute the sample mean $\bar{x}$, which estimates the unknown true mean μ of the underlying Normal distribution. We can further calculate the standard error of our estimate. Then the difference between the sample mean and the true mean divided by the standard error (which captures how many standard errors our sample mean is away from the true mean) follows a t -distribution with parameter $\nu = n - 1$. Note that the distribution is independent of μ and σ , i.e. we can compute it solely based on the degrees of freedom, which depends on the sample size!
- The t -distribution was discovered by William Sealy Gosset and published under the pseudonym Student while working at the Guinness brewery in Dublin. In his work for the brewery, he was interested in the statistical distribution of chemical properties of barley for small sample sizes and in this context he discovered the t distribution.

Student's t-test

- ▶ consider data with n realisations x_i of $X \sim \text{Norm}(\mu_0, \sigma_0)$ with unknown μ_0 and σ_0 and **sample mean**

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i$$

- ▶ for **null hypothesis** H_0 that $\mu_0 = \mu$ define **test statistic** t as

$$t := \frac{\bar{x} - \mu}{SE} = \frac{\bar{x} - \mu}{\sqrt{\frac{s^2}{n}}}$$

where $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ is (bessel-corrected) **estimate of variance**

- ▶ we have $t \sim \text{Stud}(t; n-1)$, i.e. **p-value of H_0** ($\mu_0 = \mu$) against **two-sided alternative hypothesis H_1** ($\mu_0 \neq \mu$) is

$$p := P_{H_0}(|T| \geq |t|) = 2 \cdot \int_{|t|}^{\infty} \text{Stud}(x; n-1) dx$$

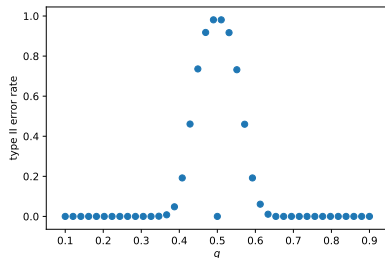
Notes:

- Why is the t -distribution helpful to test hypotheses in data? Because it is the basis for a very general test whether the mean of a sample deviates significantly from the expected value of an underlying normal distribution. This is also known as the **one-sample t -test**. The t in t -distribution is actually short for hypothesis **t**est.
- The t -test uses the fact that the difference between sample means for two samples drawn from a Normal distribution is itself normally distributed. It allows us to test the null hypothesis that a data set with sample mean $\bar{x}$ was drawn from a normal distribution with true expected value μ (i.e. $\mu_0 = \mu$) against the two-sided alternative that the true expected value of the underlying normal distribution is not μ (i.e. $\mu_0 \neq \mu$).
- The test defines the test statistic t as described above. It captures the distance of the sample mean from assumed mean μ in units of standard error of the sample mean. The standard error of the sample mean can be calculated as the Bessel-corrected estimate of variance divided by the square root of the sample size.
- Under the null hypothesis that the sample is drawn from $\text{Norm}(\mu, \sigma)$ we find that the test statistics $t \sim \text{Stud}(n - 1)$. That means, we can calculate the p -value of the null hypothesis H_0 against the two-sided alternative based on the cumulative distribution of the t -distribution. The value of t can be positive or negative (depending on whether our sample mean is larger or smaller than the true mean). Since the t -distribution is symmetric, we can use the absolute value of t and simply double the value from the cumulative distribution.

Coin tossing example

- ▶ for sufficiently large n we can approximate $\text{Binom}(n, p)$ by $\text{Norm}(np, \sqrt{np(1-p)})$ → Moivre-Laplace theorem, L09
- ▶ for observed average number $\bar{x}$ of tails in n tosses of a coin, we can **use t -test to test whether coin is fair**
- ▶ null hypothesis H_0 is that true expected number of tails is $\frac{n}{2}$, i.e. $H_0 : \mu = \frac{n}{2}$
- ▶ we test H_0 against **two-sided alternative hypothesis** that coin is biased, i.e. $H_1 : \mu \neq \frac{n}{2}$
- ▶ using test statistic $t = \frac{\bar{x} - \mu}{SE}$ the p -value of a two-sided hypothesis test can be calculated as

$$p = 2 \cdot \int_{|t|}^{\infty} \text{Stud}(x; n-1) dx$$



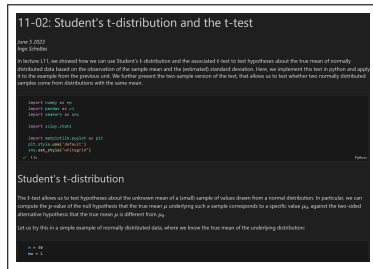
type II error rates of t -test for $n = 20$ coin tosses in 1000 tests with different values of q and $\alpha = 0.01$

Notes:

- We now apply this statistical test to an example that we already know, our coin tossing example from the beginning of the lecture. We apply one of the limit theorems introduced in lecture L09, which states that for large n and fixed p , we can approximate the binomial distribution with a Normal distribution, where the mean and variance depend on the number of tosses n and the probability p .
- With this, we can apply the t -test to our example. Our null hypothesis H_0 is that the coin is fair, i.e. $q_0 = \frac{1}{2}$ for heads. We can express this in terms of the mean of the Normal distribution that corresponds to our null hypothesis, i.e. for 20 coin tosses we have $\mu_0 = n \cdot q_0 = 10$ times heads. Different from before, we consider a two-sided alternative hypothesis H_1 , i.e. we consider both $q < \frac{1}{2}$ and $q > \frac{1}{2}$ at the same time, and thus our alternative hypothesis is that $\mu \neq \mu_0 = 10$.
- Using the sample mean $\bar{x}$, i.e. the actual number of heads observed in the data, we can now calculate the t statistic from the previous slide, and we can use the cumulative distribution function of Student's distribution to calculate the p -value of our null hypothesis that the coin is fair. We then use a significance threshold α to reject or not reject null hypothesis.
- We know that the type I error will be α , but what about the type II error. If the test actually works, we should see the type II error dropping to zero, as soon as the coin is sufficiently strongly manipulated. As the number of coin tosses n that we observe grows, the type II error will drop to values close to zero also for q close but not identical to 0.5.

Practice Session

- ▶ we **implement the t -test in python**
- ▶ we use the t -test to **test whether a sample is drawn from a Normal distribution with given mean μ**
- ▶ we apply the t -test to **test whether a coin is biased**



practice session

see notebook 12-02 in gitlab repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

Notes:

Testing linear relationships

- ▶ for estimate $\hat{\beta}_1$ we can use t -test to **test null hypothesis** H_0 **that true slope** $\beta_1 = 0$ against alternative $\beta_1 \neq 0$
- ▶ we use the **test statistic**

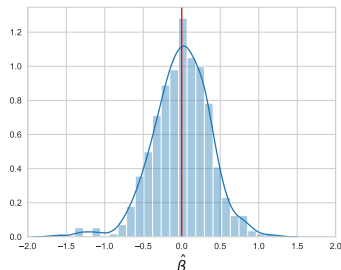
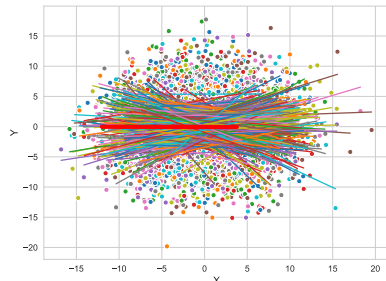
$$t = \frac{\hat{\beta}_1 - 0}{SE}$$

with standard error of linear regression estimate

$$SE := \sqrt{\frac{\frac{\text{Var}(\epsilon)}{\text{Var}(X)}}{n-2}}$$

- ▶ p -value of $H_0 (\beta_1 = 0)$ vs. $H_1 (\beta_1 \neq 0)$

$$p := P_{H_0}(|T| \geq |t|) = 2 \cdot \int_{|t|}^{\infty} \text{Stud}(x; n-1) dx$$



Notes:

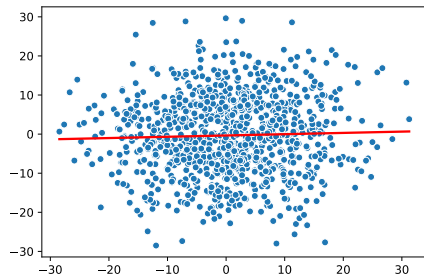
- We now have everything in place to explain how we can use Student's t -test to test linear relationships. Under the assumptions of linear regression, the estimate $\hat{\beta}_1$ of the slope of a linear model is unbiased and we obtain a Normally distributed estimate, where the true mean of many estimates corresponds to the true relationship in the data. For any null hypothesis about the true relationship, we can now use the t -test to see whether our estimate is in line with this null hypothesis or not. That is, if we want to test whether a relationship exists in the first place our null hypothesis is $\mu = \mu_0 = 0$, vs. the two-sided alternative $\mu \neq \mu_0$. We use the standard error as well as our estimate for the slope (i.e. our sample mean) to calculate the t -statistic and we then use the cumulative distribution function of Student's t -distribution to calculate a p -value.
- Small p -values indicate that it is unlikely to observe an estimate as far from zero (far, in terms of standard error of our estimate, which accounts for both the variance and the size of our data set) as in our data, if the null hypothesis that there is no linear relationship in the data is true. For large p -values we cannot reject the null hypothesis that there is no linear relationship if we do not want to risk a high type I error.

Testing linear relationships: example

example 1

1000 realisations of (X, Y) with $X \sim \text{Norm}(0, 10)$ and

► $Y \sim \text{Norm}(0, 10)$



$$\hat{\beta}_1 = 0.033$$

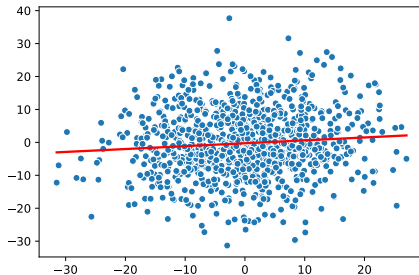
$$t = 1.045$$

$$p = 0.297$$

example 2

1000 realisations of (X, Y) with $X \sim \text{Norm}(0, 10)$ and

► $Y \sim 0.05 \cdot X + \text{Norm}(0, 10)$



$$\hat{\beta}_1 = 0.088$$

$$t = 2.759$$

$$p = 0.006$$

Notes:

- Above we apply the t -test in our two examples from before. Here, we have data with high variance, and in the first example there is no linear relationship, while there is a linear relationship in the second example. Both estimates of the slope are relatively close to zero.
- Computing the t -statistic, we can subject the hypothesis that there is no linear relationship to a formal test, finding that we can safely reject the null hypothesis in the second example (type I error probability 0.6 %), while we cannot safely reject it in the first example (type I error probability of close to 30%).

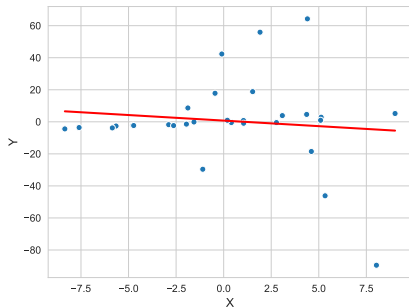
Is there a linear relationship?

- ▶ in the example we estimate $\hat{\beta}_1 = -0.69$ and t -test yields p -value of $p = 0.549$
- ▶ we find no evidence against H_0 despite linear relationship, i.e. **type II error**
- ▶ **non-normal residuals**, i.e. assumptions of t -test and linear regression not satisfied
- ▶ we need method to **test whether residuals are normally distributed**

example

50 realisations of (X, Y) with $X \sim \text{Norm}(0, 5)$ and

▶ $Y \sim 0.5 \cdot X + ?$



$$\hat{\beta}_1 = -0.69$$
$$p = 0.549$$

Notes:

- Let us continue with a more complicated example. Here, we consider a synthetic data set similar to the one that motivated the study of least squares linear regression from the likelihood perspective in lecture L09. In a moment we shall see how I have generated the data. For now it is enough to say that it contains a relatively strong linear relationship with slope 0.5. However not only does a least squares linear regression yield a wrong estimate, we also see that the t -test would not allow us to safely reject the null hypothesis that there is no linear relationship. In other words, if we apply the t -test to this example, we are likely to make a type II error, i.e. we wrongly do not reject the null hypothesis, even though it is false.
- Looking back at lecture L10, you may guess that the reason for this mistake is that, in this example, the residuals of the linear model are actually not normally distributed (which invalidates the estimate of the parameters). This implies that neither the assumptions of linear regression are fulfilled (which means our estimate is biased), nor do we meet the assumptions of the t -test (no unbiased estimate of the sample mean and no normally distributed quantities x_i).
- This brings us to another important application domain for statistical hypothesis testing: the test whether data is drawn from a Normal distribution or not. In the case of linear regression, we can use this to test the distribution of the residuals, i.e. to ensure that the assumptions of linear regression and the t -test are satisfied.

Shapiro-Wilk test

- ▶ for n independent realisations x_i of random variable X we can **test null hypothesis** H_0 that $X \sim \text{Norm}(\mu, \sigma)$
- ▶ for sample mean $\bar{x}$ compute **test statistic**

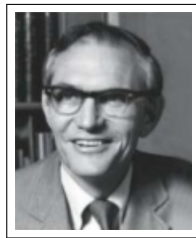
$$W := \frac{b^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where b^2 is estimate of variance of **sample from**
 $X \sim \text{Norm}(\mu, \sigma)$

- ▶ p -value of H_0 ($X \sim \text{Norm}$) against H_1 ($X \not\sim \text{Norm}$) can be calculated as

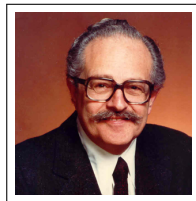
$$p := P_{H_0}(T \geq W)$$

- ▶ no closed-form distr. of W , i.e. we use **Monte-Carlo simulations or pre-computed tables**



Samuel S. Shapiro

born 1930



Martin B. Wilk

1922 – 2013

Notes:

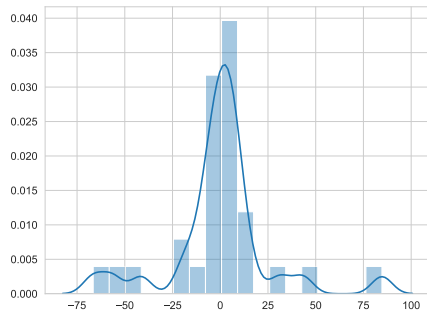
- There are different methods to test data for normality, e.g. D'Agostini's K^2 test, the Chi-Square test, or the Kolmogorov-Smirnov test. Some of those we will cover in tomorrow's practice lecture. Different tests have different advantages in terms of the power of the test (i.e. how good they are at detecting deviations from Normality) or the assumptions that they make. Here we will focus on the widely used test developed by Shapiro and Wilk. It is known for its large test power in small samples.
- We first need to formulate null and alternative hypothesis and then compute a test statistic. Here, our null hypothesis is that the data is normally distributed, while the alternative hypothesis is that it follows a different distribution.
- The test statistic W is based on the variance of a sample of size n if it was drawn from a normal distribution (in the counter) as well as an estimate for the variance of our sample (in the denominator). We can use it to calculate a p -value for H_0 . Unfortunately, there is no closed-form solution, i.e. we either have to approximate the cumulative distribution based on simulations, or we use precomputed tables that translate the sample size n and W statistic to p -values.
- The Shapiro-Wilk test has a number of disadvantages: Assuming that the data is smaller than 5000 samples it is not guaranteed to work for large data. Different from previous tests, the test logic is reversed. If we want to show that data is normally distributed we do not want to reject the null hypothesis! This changes the interpretation of type I and type II errors and we must be careful when interpreting p -values.

Normality of residuals: examples

example 1

50 realisations of (X, Y) with $X \sim \text{Norm}(0, 5)$

- ▶ $Y \sim 0.5 \cdot X + \text{Norm}(0, 1)$ for 30 realisations
- ▶ $Y \sim 0.5 \cdot X + \text{Norm}(0, 50)$ for 20 realisations



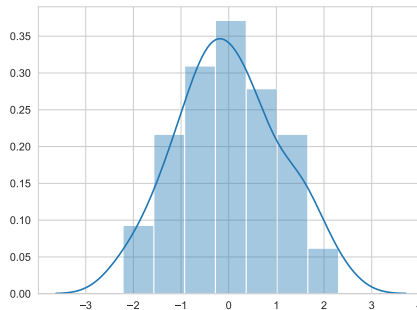
residual distribution

$$W = 0.699, p = 1.49 \cdot 10^{-6}$$

example 2

50 realisations of (X, Y) with

- ▶ $X \sim \text{Norm}(0, 5)$ and
- ▶ $Y \sim 0.5 \cdot X + \text{Norm}(0, 1)$



residual distribution

$$W = 0.986, p = 0.812$$

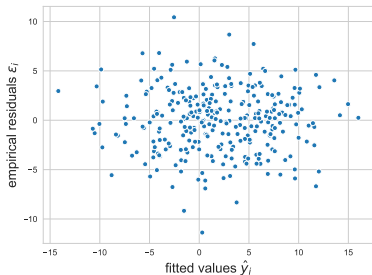
Notes:

- In the practice session, we will see how we can use `scipy`'s implementation of the Shapiro-Wilk test. In the examples above, we have applied to test the normality of the residuals for two regression examples. The first example on the left is the motivating example with non-normal residuals. This non-normality results from the way the data has been generated. In particular, we have used two different normal distributions with largely different variances. Due to the “outliers”, the residuals of the combined data do not follow a normal distribution. The Shapiro-Wilk test detects this, as expressed by the small p -value.
- The opposite case is shown in the example on the right, where the residuals follow a Normal distribution. Consequently, the large p -value calculated using the W statistic does not allow us to reject the null hypothesis that residuals are normally distributed.

we can **graphically check for violations of assumptions of least squares regression**

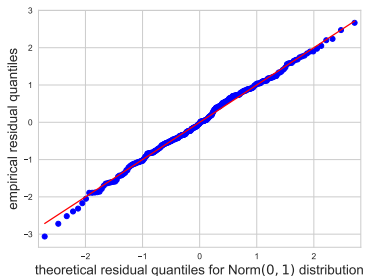
Tukey-Anscombe plot

plot residuals ϵ_i (y-axis) of a linear model against predicted values $\hat{y}_i$ of dependent variable (x-axis)



Quantile-Quantile (QQ) plot

plot empirical quantiles of residuals (y-axis) against theoretical quantiles of Normal distribution (x-axis)



example 1

assumptions of linear regression seem to be **fulfilled**

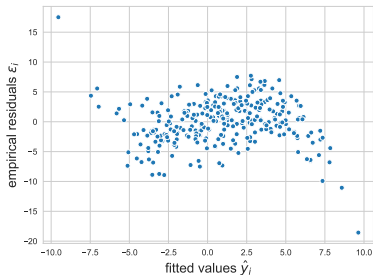
Notes:

- To conclude our discussion of **regression diagnostics**, we finally introduce two frequently used graphical techniques to check for violations of the assumptions of ordinary least squares regression. Those techniques are based on a visual inspection of patterns in specific types of plots.
- The first method is based on the so-called **Tukey-Anscombe plot**, which plots the empirical residuals $\epsilon_i := y_i - \hat{y}_i$ of a linear model against the predicted values $\hat{y}_i$ of the dependent variable. If the residual distribution is homoskedastic and centred around zero, we should see random fluctuations around zero in the Tukey-Anscombe plot, and there should not be any dependency of the residuals on the predicted dependent variables. This is the case in the example above (see left figure), where the assumptions of linear regression are fulfilled.
- The second method is based on the so-called **Quantile-Quantile or QQ-plot**, which plots the theoretical quantiles of normally distributed residuals against the empirical residual quantiles. If the residuals are distributed according to a Normal distribution, we expect a perfect linear relationship between the theoretical and empirical quantiles of the residual distribution, where the slope and intercept depend on the mean and variance of the underlying Normal distribution. This is the case in the example above (see right figure).

we can **graphically check for violations of assumptions of ordinary least squares regression**

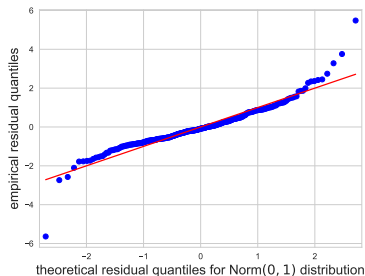
Tukey-Anscombe plot

plot residuals ϵ_i (y-axis) of a linear model against predicted values $\hat{y}_i$ of dependent variable (x-axis)



Quantile-Quantile (QQ) plot

plot empirical quantiles of residuals (y-axis) against theoretical quantiles of Normal distribution (x-axis)



example 2

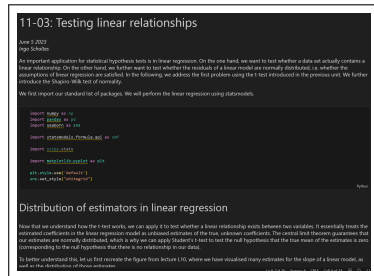
assumptions of linear regression are **violated**

Notes:

- We now repeat those two plots in an example, where the assumptions of linear regression are violated. Here we have used a data set with a non-linear relationship, which results in non-normal and heteroskedastic residuals.
- The heteroskedasticity of residuals is indicated by the presence of a strong non-random pattern in the Tukey-Anscombe plot. This is likely to result in a non-zero expected residual (which translates to a systematic bias in the parameter estimation). We could potentially fix this based on a non-linear transformation of the response variable (log, square-root, polynomial transformation), after which we check whether the pattern in the Tukey-Anscombe plot disappears.
- The non-normality of residuals in this example is further shown by a non-linear pattern in the Quantile-Quantile plot. Again, we could repeat this plot after a reasonable transformation of the dependent variable has been chosen.

Practice Session

- ▶ we use python to **test data for linear relationships**
- ▶ we apply the t -test to **test whether a linear relationship exists in a given data set**
- ▶ we use the Shapiro-Wilk test to test for the **normality of residuals**
- ▶ We use graphical methods for **regression diagnostics**



practice session

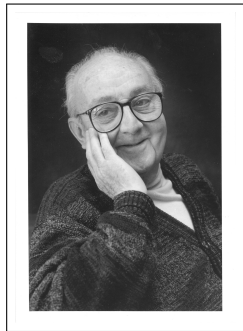
see notebook 12-03 in gitlab repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

Notes:

Conclusion

- ▶ we explored **statistical methods to test hypotheses**
- ▶ null hypothesis corresponds to **theory or statistical proposition that we seek to falsify**
- ▶ p -value = belief in null hypothesis vs. alternative
- ▶ significance threshold α determines **type-I error rate**, i.e. rate at which we **wrongly reject null hypothesis**
- ▶ choice of **alternative hypothesis, test statistic, and α** is crucial to minimise **type-II error rate**
- ▶ statistical inference & hypothesis testing are foundation of **probabilistic reasoning** in empirical sciences
- ▶ important basis to understand **limitations and fallacies of probabilistic reasoning in data science** → L12



George Box, 1919 – 2013

“the statistician knows [...] that in nature there never was a normal distribution, there never was a straight line, yet with normal and linear assumptions, known to be false, he can often derive results which match, to a useful approximation, those found in the real world.”

→ George Box, JASA, Vol. 71, pp. 791-799, 1976

Notes:

Self-study questions

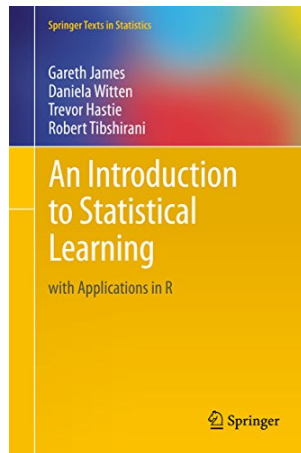
1. Explain the role of null hypothesis and alternative hypothesis in statistical hypothesis testing.
2. Why is it wrong if we say that we proved an alternative hypothesis?
3. Given an example for type I and type II errors in hypothesis testing.
4. How does the choice of alternative hypotheses affect the type I or type II error rate?
5. How does the choice of the significance threshold affect the type I or type II error rate?
6. Write down a probabilistic interpretation of the p -value.
7. Write down the t -statistic and explain how we can use it to test linear relationships.
8. Explain the difference between the statistical inference of the parameters of a linear regression model and the statistical testing of a linear relationship.
9. How can we test whether the residuals of a linear model are normally distributed?
10. What is heteroskedasticity and how can we assess it in linear regression?
11. Explain the difference between a QQ- and a Tukey-Anscombe plot.

Notes:

References

reading list

- ▶ Christopher M. Bishop: **Pattern Recognition and Machine Learning**, Springer, 2006 → [Section 2.3](#)
- ▶ T. Hastie et al.: **The Elements of Statistical Learning**, Springer, 2009 → [Chapters 3 and 7](#)
- ▶ Werner A. Stahel: **Statistische Datenanalyse**, Vieweg und Teubner, 2008 → [Chapter 13](#)
- ▶ P. Bühlmann, M. Mächler: **Computational Statistics**, lecture notes, 2012, → [Chapter 1](#)
- ▶ Norman Draper and Harry Smith: **Applied Regression Analysis**, Wiley, 1998 → [Chapter 10](#)



Notes: